Body Leveling in Wheel-Legged Obstacle Traversal with Residual PPO

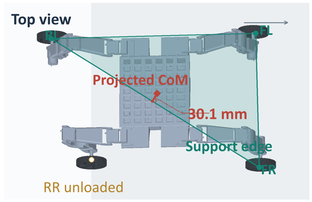
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1 Introduction

Wheel-legged robots combine efficient rolling with wheel lifting to cross steps [1,2]. As each wheel unloads and returns to contact, the body can tilt and the available support changes. This project targets a more level body during obstacle traversal. Body inclination measures how far the chassis departs from a horizontal posture.

We compare Recorded Replay and Residual PPO on 50 and 100 mm steps. Residual PPO combines recorded motion with learned joint and wheel corrections. Contact and support checks guide stage progression and corrective control. Matched distance–friction tests compare body inclination in paired successful traversals and report completion across all conditions. We assess roll and pitch together with total inclination, while projected CoM position describes geometric support.

Body leveling and geometric support



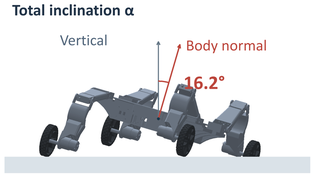


Figure 1. 50 mm Replay at 59.258 s. The projected centre of mass (CoM) is 30.1 mm from the support boundary. α is the angle between the body normal and vertical [5].

2 Research Question and Objectives

Can residual PPO keep the body more level during step traversal with contact and support checks?

- Reduce body inclination during successful step crossings.
- Assess body leveling alongside geometric support.
- Compare posture and completion across 39 conditions per height.

3 System Overview

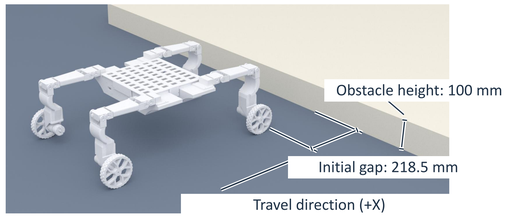


Figure 2(a). Reference scene with a 100 mm step and a 218.5 mm initial gap. The study compares 50 and 100 mm steps, with travel along +X.

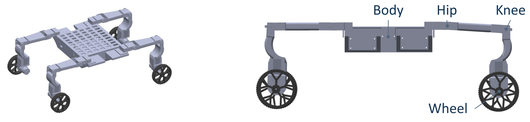


Figure 2(b). Robot anatomy in isometric and side views.

Robot model and observed state

System specification	Value or recorded quantity
Simulation platform	Isaac Sim + Isaac Lab (PhysX)
Robot mass and bodies	2.931 kg; 13 rigid bodies
Actuation	8 leg joints; 4 driven wheels
Model envelope	660 × 441 × 225 mm (authored pose)
Wheel radius	Approximately 50 mm
Physics and observation	120 Hz
Recorded state	Body orientation; joints/wheels; body masses/poses; bearing contacts; wheel/platform geometry

Table 1. Robot model, simulation platform and recorded quantities.

4 Methods

Recorded Replay

- Read demonstrated joint and wheel commands from the recorded sequence.
- Map joint targets to the robot and use joint feedback for servo tracking.
- Execute the recorded sequence through the actuator interface and physical limits.

Residual PPO

- Use measured body pose, joint state and contacts to compute allowed corrections.
- Construct nominal targets from the recording and predict residuals from the observation.
- Bound the corrections, combine them with nominal targets, then apply coordination and motor limits.

Both methods command 8 leg angles and 4 wheel speeds through the same 120 Hz interface.

5 Residual PPO

50 mm: residual PPO training

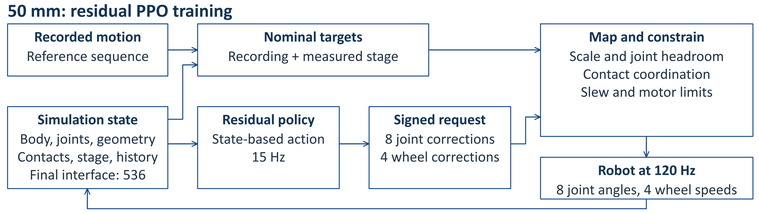


Figure 3. The controller combines recorded motion with learned residual corrections. Contact and support checks guide stage progression and corrective control. Joint and wheel commands pass through actuator limits before execution.

$$u_t = C(u_t^{nom} + \Delta u_t, o_t)$$

u_t^{nom} is the nominal target based on recorded motion. Δu_t is the learned correction. C coordinates contact and applies actuator limits using the current state o_t.

$$L(\psi) = E_t[\min(\rho_t \hat{A}_t, \mathrm{clip}(\rho_t, 1-\varepsilon, 1+\varepsilon)\hat{A}_t)]$$

PPO favors actions with higher estimated return [4]. A critic and GAE estimate advantage $\hat{A}_t$. The probability ratio ρ_t compares the new and previous policies, with clipping parameter $\varepsilon = 0.2$.

PPO training cycle

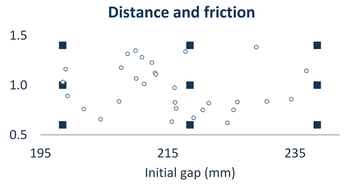
Collect rollouts → Estimate advantage → Clipped PPO update → Policy evaluation

Isaac Sim 5.1, 50 mm step. Selected early rollouts: 128 decisions; later local training: 1 environment × 512 decisions/rollout. PPO: 5 epochs, 4 minibatches, Adam with adaptive learning rate (batch-mean KL target 0.01). Later $\gamma = 0.9985$; GAE $\lambda = 0.99$.

Training component	Reward design
Early motion	Reward task progress and add normalized roll/pitch and angular-rate costs (weight 0.03), then body–obstacle gap costs below 20 mm (combined weight 0.06).
Rear-right transfer	Add one-time events: RR touchdown +10, RL lift +10, crossing +15 and placement +20. Completion +80, failure −40 and time −0.02/s. Potential-based progress shaping: weight 5.
Following transfer	Reward front-leg preparation and support transfer toward front-right, followed by rear-left lift and placement. Retain completion and time terms.
Front-left refinement	Reward forward travel +5/m. Penalize reverse wheel commands and loaded counter-rotation without progress (each up to 0.1/s). Include the completion return.
Completion objective	Combine forward travel (0.35), leg-pose recovery (0.40) and settling (0.25) in the completion potential.

Table 2. Reward emphasis changes across training stages. Completion terms also contribute to front-left training.

6 Experimental Design



Test procedure

- Reset each method independently at the same gap and friction.
- Run each height to physical completion or the 200 s limit.
- Record body pose, bearing contacts and progress at 120 Hz.
- Compare completion across 39 conditions per height. Assess posture over paired completions.

Figure 4. Matched design: 9 fixed and 30 sampled conditions at each height. Independent resets and a 200 s limit define each trial.

7 Results

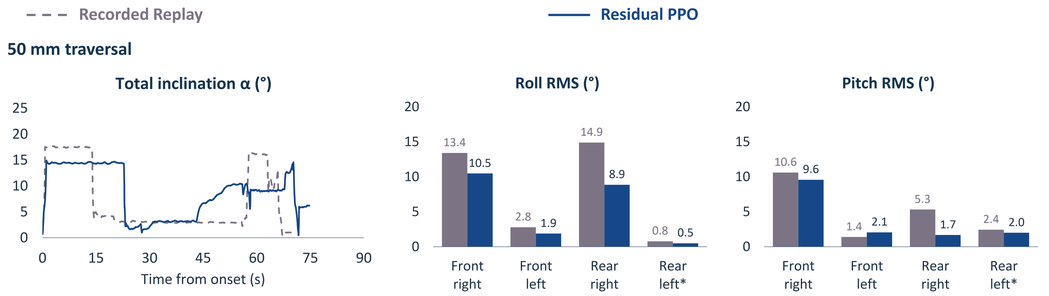
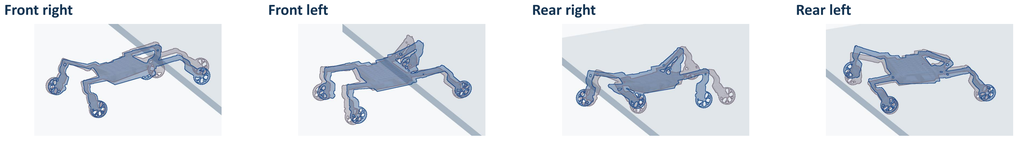


Figure 5. 50 mm: reference curve and time-weighted first-lift RMS. *Rear-left PPO bars are author-reported; raw traces unavailable. 100 mm: joint residual policy with learned wheel gains. The nominal curve shows 15 Hz poses up to geometric completion. Bars use 120 Hz RMS over identical fixed Replay windows. Front-right pools two transfers.

50 mm body inclination at corresponding wheel crossings



100 mm updated nominal test: body inclination at selected traversal stages

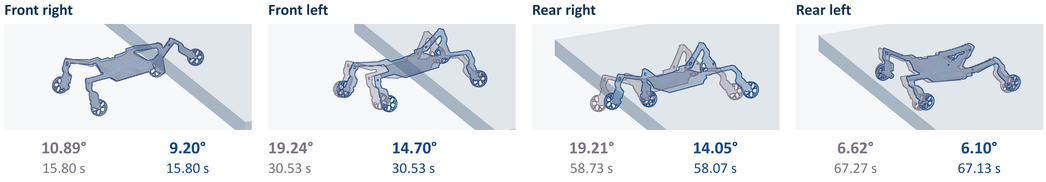


Figure 6. 50 mm prior reference and 100 mm updated nominal test. Wheel-centre crossing poses at each method's time, except 100 mm FL: both methods at t = 30.53 s, after transfer. Actual retained 15 Hz poses for 100 mm. Matched camera and scale. Muted purple: Replay. Blue: PPO.

Completion and traversal performance

Height	Method	Completed	Peak α*	Time*
50 mm	Recorded Replay	39 / 39	17.74°	70.15 s
50 mm	Residual PPO	28 / 39	15.01°	74.35 s
100 mm	Recorded Replay	31 / 39	19.41°	83.79 s
100 mm	Updated PPO	37 / 39	23.43°	83.76 s

Table 3. Common completion across 39 conditions (100 mm: known regression set). *Medians of 28/30 paired common successes at 50/100 mm. Time ends at verified common completion. The 100 mm peak α ends at geometric completion.

Project materials
Videos and source data

8 Conclusion

We developed residual PPO control for body leveling during wheel-legged obstacle traversal. The controller combines recorded motion with state-based joint and wheel corrections, while contact and support checks guide execution. Matched distance and friction tests assess body posture, geometric support and completion together.

At 50 mm, PPO reduced peak inclination in 27 of 28 paired completed trials. In the 100 mm regression tests, tilt RMS through geometric completion fell by a mean 16.7% versus Replay across 25 paired full successes. Full completion, requiring wheel placement and 0.5 s of stability, reached 32/39 compared with Replay's 27/39.

Future work will improve completion reliability and peak-inclination control across unseen terrain and friction conditions. We will study how reward design balances traversal progress, body posture and support, repeat training across seeds and validate the controller on physical hardware.

[1] Bjelonic et al. Keep Rollin'. IEEE RA-L, 2019. [2] Chamorro et al. Blind Stair Climbing. ICRA, 2024.
[3] Johannink et al. Residual Reinforcement Learning for Robot Control. ICRA, 2019.
[4] Schulman et al. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.
[5] Caron et al. ZMP Support Areas for Multi-contact Mobility. IEEE T-RO, 2017.